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Australian Antarctic Division
Owners Engineer – Asbestos Removal Programme

Report

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ABBREVIATIONS

The following summarises terms, abbreviations, and acronyms used within this document.

AAD	Australian Antarctic Division
ACM	Asbestos Containing Material
AMP	Asbestos Management Plan
ARP	Asbestos Removal Programme
FY	Financial Year
PPE	Personal Protective Equipment
WHS	Work Health and Safety

1 INTRODUCTION

1.1 BACKGROUND

The Australian Antarctic Division (AAD) manages asbestos-containing materials (ACMs) across its Antarctic and sub-Antarctic stations through an established framework consisting of the Asbestos Management Plan (AMP) and station-specific Asbestos Registers. While these documents provide the basis for identifying, assessing and managing ACMs, they do not provide a structured framework for determining when asbestos should be removed, how removal activities should be prioritised, or the likely cost and programme implications of those decisions.

This Asbestos Removal Programme (ARP) has therefore been developed to support long-term planning by establishing a consistent methodology for assessing, prioritising, scheduling and forecasting asbestos removal works across AAD facilities over a 10-year planning horizon. The programme is intended to complement, not replace, the existing AMP and Asbestos Registers, and focuses on strategic decision-making, programme development and cost planning rather than the detailed execution requirements for individual asbestos removal activities.

1.2 SCOPE

This Asbestos Removal Programme applies to asbestos-containing materials (ACMs) identified within the current Asbestos Registers for AAD Casey, Davis, Macquarie Island and Mawson stations and establishes a framework for assessing, prioritising, scheduling and forecasting asbestos removal activities over a 10-year planning period.

The scope includes the development of a prioritisation methodology, programme planning assumptions, indicative removal and reinstatement requirements, scheduling considerations, constraints assessment, and high-level cost forecasting to support long-term asset and maintenance planning.

The programme utilises information from the Asbestos Management Plan and Asbestos Registers as key inputs but does not replace or supersede these documents. Detailed asbestos removal methodologies, contractor-specific removal control plans, regulatory notifications, air monitoring, decontamination procedures, waste disposal arrangements, clearance inspections and other activity-specific execution requirements are outside the scope of this programme and shall be developed as part of individual removal work packages prior to the commencement of works.

2 EXISTING MANAGEMENT

2.1 COMPLIANCE

The Asbestos Management Plan (AMP) has been developed by the AAD in line with national Work Health and Safety (WHS) Regulations and the Codes of Practice for asbestos. The ARP has been developed in accordance with the AMP, using it as the primary reference to ensure alignment with its policies.

2.2 ASBESTOS MANAGEMENT PLAN

The Asbestos Management Plan for the Australian Antarctic Territory and Macquarie Island is a key tool in managing all instances of asbestos containing material. It provides guidance on how to protect the health and safety of employees and expeditioners following the identification of ACMs in the workplace.

The ARP is not intended to redefine or supersede the AMP. The ARP will focus on how to prioritise and schedule the decisions made by AAD regarding ACM removal from site and will refer to existing policies and assessments contained within the AMP as required.

Where removal of ACM is not reasonably practicable or preferable, the ACMs can be protected or enclosed, sealed or encapsulated, or repaired as directed by the AMP or noted in the Asbestos Register.

2.3 ASBESTOS REGISTERS

The following Asbestos Registers have been provided for WSP review and inclusion in the tool:

- Asbestos Register – Casey Station (Version 5, 16/06/2021)
- Asbestos Register – Davis Station (Version 5, 16/06/2021)
- Asbestos Register – Macquarie Island Station (Version 9, 27/02/2025)
- Asbestos Register – Mawson Station (Version 5, 16/01/2023)

2.4 ARP INTERFACE WITH EXISTING REGISTERS

The ARP is intended to operate in conjunction with the existing Asbestos Management Plan (AMP) and station-specific Asbestos Registers. The AMP remains the governing document for the identification, assessment and ongoing management of asbestos-containing materials (ACMs), while the Asbestos Registers remain the primary source of information on ACM locations, condition assessments, inspection records and management actions. The ARP uses information from these documents as key inputs to support the prioritisation, scheduling and cost forecasting of future asbestos removal activities.

The ARP should always be read in conjunction with the relevant asbestos register for the latest observation records and comments.

Where information in the programme tool differs from the current approved Asbestos Register, the current Asbestos Register shall take precedence for ACM identification, condition and inspection information. The programme tool shall take precedence only for programme-specific decisions, assumptions, timing and cost forecasts.

3 PROGRAMME OBJECTIVES

3.1 PROGRAMME OBJECTIVES

The objectives of this programme are to:

- Develop a structured methodology for the prioritisation of asbestos removal works.
- Establish a consistent basis for evaluating removal requirements across AAD stations.
- Integrate asbestos removal activities with planned maintenance, refurbishment, demolition and capital works programmes.
- Identify the indicative scope, timing and delivery considerations associated with future asbestos removal works.
- Record programme assumptions, constraints, dependencies and decision-making criteria.
- Forecast removal and reinstatement costs to support long-term asset and financial planning.

3.2 PROGRAMME DURATION

Through discussions between AAD and WSP regarding the objective of achieving complete removal of all ACM from Antarctic and sub-Antarctic stations within a 10-year timeframe, a shared view was developed that this target is not considered realistic or appropriate. This mutual conclusion was reached considering the following key constraints:

- The potential impact of an accelerated removal programme on station operations.
- The practical challenges associated with accommodating and mobilising a large workforce to these remote stations.
- The significant costs associated with removal of asbestos in remote regions.
- A 10-year target may not represent the preferred whole-of-life approach, particularly where demolition, replacement, or major maintenance activities are anticipated outside the 10-year planning timeframe.
- The risk associated with an extended time to removal can be readily managed.

The ARP maintains a recorded planning timeframe of 10-years, noting this typically exceeds the majority of AAD's forecasted project planning. Where ACM removal is pushed beyond 10-years, justification shall be recorded in the register.

3.3 PROGRAMME DEVELOPMENT APPROACH

[commentary to be added regarding the best approach at balancing removal tasks across seasons and across stations. Best done when the scale of work required is completed from other stations]

3.4 REMOVAL STRATEGY

The AAD asbestos removal strategy is based on the principle of planned, risk-informed and opportunity-driven removal. While ACMs that present an elevated risk may require prioritised action, the complete removal of all ACMs is not always the most practical, efficient or risk-appropriate outcome. Where ACMs remain in good condition and can be safely managed in accordance with the Asbestos Management Plan (AMP), continued management may be preferable until a suitable removal opportunity arises.

The programme seeks to progressively reduce the AAD asbestos liability by prioritising removal activities that deliver the greatest reduction in risk, operational burden and future maintenance requirements. Asbestos removal will be coordinated with planned maintenance, refurbishment works, building upgrades, demolitions and other asset interventions wherever practicable to minimise duplication of effort, reduce disruption to station operations and optimise the use of limited logistical and financial resources.

Given the unique operational environment of Antarctic and sub-Antarctic facilities, removal decisions must also consider seasonal access windows, freight requirements, accommodation capacity, trade availability, station operational priorities and wider project interfaces. Accordingly, the ARP adopts a long-term planning approach that balances risk reduction with practical delivery considerations to establish an achievable and cost-effective programme of works.

The removal strategy recognises that asbestos management is an ongoing process and that removal priorities may change over time as asset conditions, operational requirements, project plans and asbestos register information are updated. The programme should therefore be viewed as a dynamic planning tool that supports informed decision-making and periodic review rather than a fixed commitment to undertake specific works in a particular year.

3.5 DECISION-MAKING PRINCIPLES

The following principles have been adopted to guide asbestos removal decisions and programme development:

- **Protect health and safety**
  The protection of personnel, contractors, visitors and the environment shall remain the primary consideration when assessing asbestos removal priorities. Where ACMs present an elevated risk of disturbance or exposure, timely action shall be considered.
- **Prioritise removal where risk reduction is meaningful**
  Removal should focus on ACMs where it will achieve a demonstrable reduction in risk, management burden, maintenance constraints or future operational impacts.
- **Coordinate removal with planned works**
  Where practicable, asbestos removal should be aligned with maintenance activities, refurbishment projects, building upgrades or demolitions to maximise efficiency and minimise repeat disruptions.
- **Avoid unnecessary disturbance of ACMs**
  Removal should not be undertaken solely because asbestos is present. Consideration should be given to whether continued management, encapsulation or enclosure remains an appropriate and lower-risk option until a suitable removal opportunity exists.
- **Consider whole-of-life outcomes**
  Decisions should account for both the immediate removal works and any associated reinstatement, maintenance, operational and lifecycle implications.
- **Recognise logistical and operational constraints**
  Programme priorities shall consider the practical realities of Antarctic operations, including seasonal access, freight limitations, accommodation capacity, workforce availability and station operational requirements.
- **Maintain transparency and traceability**
  Key assumptions, dependencies, drivers, constraints and scheduling decisions should be recorded to ensure programme recommendations can be understood, reviewed and updated over time.
- **Maintain programme flexibility**
  The ARP shall be reviewed and updated as new information becomes available, recognising that

priorities may change in response to updated register information, asset strategies, project requirements or operational needs.

3.6 KEY ASSUMPTIONS

The following assumptions have been made in the development of the ARP and the supplementary planning and costing tool:

- The programme assumes a five-day working week and an eight-hour working day, with four productive tool hours per day.
- Removal labour allowances are based on two AAD-provided Casey Station worked examples, normalised in the tool using the complexity and accessibility adjusters. There are then used to pro-rata labour required for other ACM removal tasks.
- The reinstatement labour rates provided by North Projects Pty Ltd are assumed to represent productive tool time.
- Freight volumes include bulking allowances for removed ACM, reinstatement materials, PPE and basic removal equipment.

4 PRIORITISATION

4.1 PRIORITISATION OVERVIEW

The decision guidance is the primary tool in assisting the AAD with prioritising the removal of asbestos containing materials. The approach for each registered ACM is to work through the known future of the item. Where none are known, providing simple methods to compare or identify potential ACM removal tasks. Figure 1 provides a high-level summary of the ACM removal decision filters to consider in allocating a removal season.

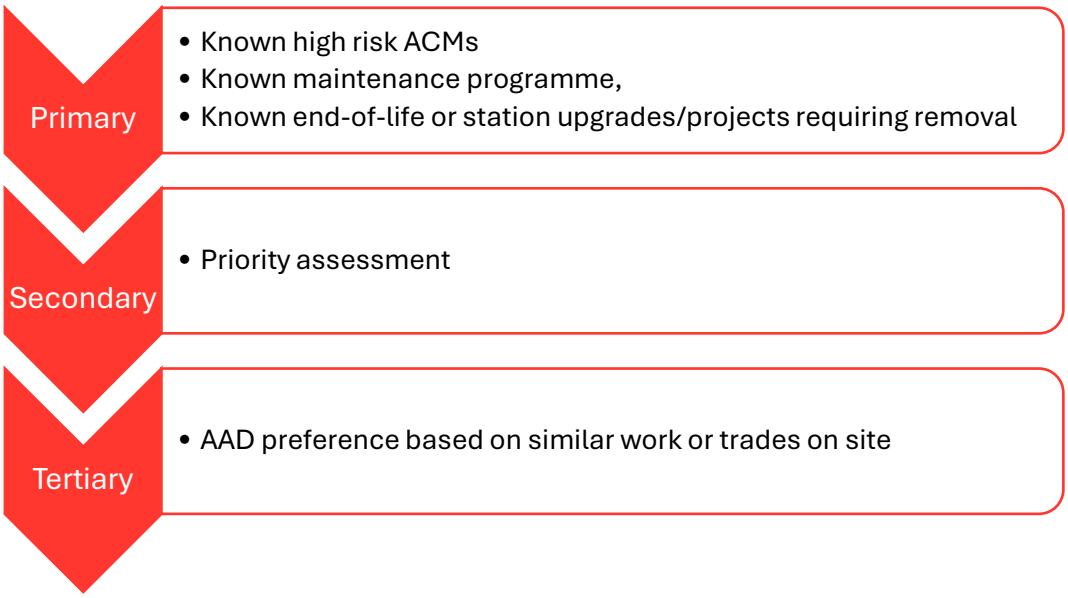


Figure 1 – ACM removal task decision filtering

The programme tool will provide a means of recording the decision for ACM removal and the assumptions around the decision that is reached. With the tool in-part being used as an assumption register to track the ownership and decisions for planned ACM removal.

4.2 PRIMARY PRIORITISATION

4.2.1 HIGH RISK ASBESTOS

The existing Asbestos Registers have completed material risk assessments for identified ACMs. Where existing or newly input ACM items in the register report a material risk which would typically be considered outside AAD's risk appetite and therefore actioned for urgent removal, the item shall be included in the programme for removal within the coming seasons. Based on existing Asbestos Register material risk assessments, a risk score of 1 is considered high and therefore requiring rectification in the next available season.

4.2.2 END-OF-LIFE

Where a structure containing ACMs is planned for demolition or replacement due to nearing end-of-life the requirement for ACM removal ahead of this time should be captured in the programme.

As the programme addresses a 10-year period for ACM removal, any structures that have an end-of-life, and therefore planned demolition within the next 10-years, should be added to the programme. Where a known end-of-life date is outside the 10-year programme, it should be recorded in the tool as this will factor into future

updates of the programme. The ACM item should then be assessed against other decision filters to identify if there is another requirement for removal or if the preferred approach is to continue monitoring/encapsulation as per the existing AMP.

For planning purposes, ACM in structures scheduled to reach end of life within the next 10 years should generally be removed in the season before demolition, enabling safe demolition the following season. For buildings containing services critical to station operations, such as plant rooms, ACM removal and demolition are assumed to occur in the same season.

4.2.3 MAINTENANCE

If an ACM is scheduled to be removed as part of planned maintenance, it shall be allocated the season of planned maintenance work for removal. For instance, ACM gasket replacements would be expected to be removed the year the plant being replaced or maintained.

Where an ACM is expected to be disturbed as part of access for maintenance (e.g. removal of ACM containing access panel to access services), its need for removal shall be considered ahead of the scheduled maintenance. Consideration for removal should be made using the existing risk assessment in the current asbestos register while also considering the assumed effort required for full ACM removal. For example, it might not be reasonable to consider removing a large volume of low-risk ACM interior wall lining to access a valve that only requires maintenance every 10 years.

4.2.4 PLANNED STATION UPGRADES/PROJECTS

Where AAD has planned station upgrades involving alterations, additions or replacement of a structure containing ACMs, the project scope should consider whether ACMs within the affected structure should be removed as part of the works. For planning purposes, ACM removal should generally be scheduled for the season before the upgrade works are expected to commence.

Where known, AAD should also consider Alliance-planned projects that may affect structures containing ACMs. These projects should be treated consistently with AAD-planned station upgrades, with ACM removal generally scheduled at least one season before the associated upgrade works are expected to commence.

4.3 SECONDARY PRIORITISATION

4.3.1 PRIORITY ASSESSMENT

A useful secondary means of assisting prioritisation of ACM removal is the priority assessment. The approach adopted aligns with the international best practice for managing asbestos which has been used to inform similar methods for risk assessment and prioritisation within Australia. The approach requires the material risk score from the asbestos registers to be combined with the priority assessment score. The final score is then used for comparison between ACM items.

The material assessment contained within the existing asbestos registers identifies higher-risk ACMs, including materials that may more readily release airborne fibres if disturbed. However, the material assessment score alone does not determine removal priority. Priority for remedial action should be confirmed through a risk-based priority assessment that considers additional factors such as:

- Maintenance activity
- Occupant activity
- Likelihood of disturbance
- Human exposure potential

Combining the material and priority assessments provides a basis for comparing ACMs by considering both fibre release potential and the need for remedial action due to exposure to people. The combined assessment results can then be used to prioritise ACMs for removal. Where removal is not reasonably practicable, ACMs may be protected or monitored in accordance with the existing asbestos management plan.

As part of the ARP development, the AAD have carried out a revised material assessment combined with a new building use assessment to develop new weighted risk-based scores for their identified ACMs. The output increases the risk range the 4 (low risk) to 1 (high risk) approach used in the existing registers to a 21% (low risk) to 100% (high risk) scoring. This provides greater resolution when comparing priorities for removal between two ACM items.

Lower scores will represent ACMs with poorer material condition, greater degradation or friability, and higher likelihood for disturbance due to the use of the space. These items should generally be ranked ahead of ACMs with higher scores. Where scores are similar, for example a high-risk material in a remote plant room compared with a low-risk item in a regularly occupied space, comparison comes down to the planning preference of AAD and tertiary considerations.

4.4 TERTIARY DECISION FILTERS

Where the primary and secondary prioritisation filters do not identify a clear removal season, tertiary decision filters should be applied to determine the most practical removal opportunity. Some examples of tertiary decision considerations include:

- **Potential labour efficiencies**
  Consideration of the primary trade planned for removal works. It may be AAD's preference to keep removal tasks assigned to the same trade in an individual season or split them across different trades.
- **Package removal activities**
  Consider whether the work can be combined with other removal, maintenance or project activities planned at the station to reduce mobilisation and disruption.
- **Seasonal suitability**
  Consider whether the removal is internal or external and whether it is better suited to summer or winter delivery.
- **Station access and accommodation**
  Consider station access, accommodation and bed capacity when allocating the removal season.
- **Freight availability**
  Consider whether sufficient freight capacity is available for removal equipment, reinstatement materials and returned ACM waste.

4.5 ACM REMOVAL BEYOND THE PROGRAMME

Some ACM removal activities may fall outside the 10-year programme where they are assessed as lower priority, remain in a stable and manageable condition, or are linked to asset renewal or major works planned beyond the programme period. These items should remain subject to the existing Asbestos Management Plan and Asbestos Registers, with their timing reconsidered if condition, accessibility, operational requirements, or planned works change.

Below is a summary of the removal tasks or structures that should typically be removed from the ARP scope or pushed beyond the 10-year programme with ongoing management as per the existing Asbestos Management Plan:

- **Heritage structures**
  Heritage listed structures or structures where removal of ACM will have a risk on the structural integrity of the structure should be excluded from the ARP.
- **Huts**
  AAD has advised that huts should be excluded from the ARP due to their remoteness making safe removal and logistics complex. Many of the huts are also heritage listed structures and/or structurally sensitive to ACM removal activities.
- **Gaskets**
  Gaskets should remain outside the 10-year programme until their replacement timing is confirmed through AAD's maintenance programme, avoiding unnecessary disruption to live services. Their exclusion is not expected to materially affect programme costs due to the limited labour and cost associated with removal.
- **Wilkes Station**
  The Wilkes Station (near Casey Station) is an abandoned site that is excluded from the ARP and will be managed as part of a future removal programme.
- **Floor sheeting**
  Floor sheets are typically installed such that only partial ACM removal can be achieved without full demolition of the structure. Where flooring is in good condition and unlikely to be disturbed, it may remain in place and managed using the AMP. Complete removal may require extensive strip-out or reconstruction; therefore, partial removal during refurbishment may be appropriate where inaccessible material, including beneath floor plates, can be safely retained.
- [AAD to confirm any further items to list here]

5 REMOVAL & REINSTATEMENT

5.1 REMOVAL SCOPE

The removal scope is established for each ACM item by considering how the material is used, the extent that can reasonably be accessed, and whether removal is linked to maintenance, refurbishment, demolition or another planned project. Items such as gaskets, hatches, access panels and components associated with plant or services are generally linked to the maintenance or replacement of the associated asset. Where no maintenance timing has been confirmed, these items remain managed under the AMP rather than being assigned an arbitrary removal season.

For linings, flooring, cladding, insulation, debris and other distributed materials, the proposed treatment also considers whether the full extent can be removed without significant impact on station operations, structural elements, services or adjacent construction. The ACM item, material type, location, estimated quantity and accessibility are used to define and cost the removal task, including the likely removal effort and operational impact, without prescribing the contractor's removal methodology. Table 1 summarises the treatment types used to define the intended removal outcome for each item.

Table 1 – Treatment types for asbestos-containing material items

Treatment type	Description	Example
Complete	The item has already been addressed and no further testing, removal or management action is required under the programme.	An ACM item that has previously been removed and closed out.
Test Required	The item is suspected to contain asbestos but has not been confirmed. Sampling and testing are required before the treatment and removal scope can be confirmed.	An item recorded in the programme tool as suspected ACM without laboratory confirmation.
Removal – Full	All accessible ACM can be removed without materially affecting station operations, adjacent building systems or structural elements.	Accessible wall lining that can be removed in full without disrupting plant room operation or structural steelwork.
Removal – Partial	Only part of the ACM can reasonably be removed because some material is inaccessible, concealed, built over or cannot be removed without substantial impact on other building elements. Remaining ACM is retained and managed under the AMP.	Flooring beneath internal partitions, where exposed areas can be removed but material beneath the partitions must remain.
Manage per AMP	The item remains managed under the Asbestos Management Plan and has not been assigned a removal season. Treatment is linked to future maintenance, refurbishment, replacement or end-of-life works.	Hatches or gaskets retained until the associated component is maintained or replaced.

5.2 REINSTATEMENT SCOPE

The reinstatement scope is determined by whether the item being removed performs an ongoing function, forms part of the building fabric or is required to maintain the operation of an asset. Items such as wall and floor linings, gaskets, insulation and other functional components are assumed to be reinstated using a modern asbestos-free equivalent with similar performance. The reinstatement quantity is based on the area or extent of material removed, with an allowance for packaging and bulking applied when estimating freight to station. Where current replacement products are supplied in a greater thickness than the existing material, such as modern wall linings, flooring or insulation, the minimum commercially available thickness is used to

estimate the reinstatement material volume and associated freight cost. Reinstatement is generally not required where ACM is removed as part of the end-of-life demolition or full replacement of a structure. For planning purposes, ACM should generally be removed in the season before demolition, enabling safe demolition in the following season. For structures containing services critical to station operations, such as plant rooms, ACM removal and demolition are assumed to occur in the same season.

6 COST MANAGEMENT

6.1 ESTIMATION METHODOLOGY

The cost model is based on the following model:

$$Total\ Cost = (\,[Freight] + [Materials] + [Labour\ x\ Cost\ Adjuster]) \times [Inflation\ Adjuster]$$

6.1.1 REMOVAL LABOUR ESTIMATION

The rate of ACM removal is difficult to quantify because of the unique nature of delivering asbestos works in Antarctica. Removal productivity can be influenced by the limited work windows, remote logistics, containment and waste-handling requirements, and the need to complete work around station operations.

As limited comparable project data is available, and bottom up estimates would be time consuming and to detailed for a 10-year programme, a high-level baseline was developed using the two removal examples provided by AAD, based on the recorded volume of ACM removed and the labour taken to complete the work. Those projects selected include the Red Shed duct board removal completed over the 2025/26 season and the Old Trades (Chippies) Workshop completed over the 2018/19 season.

The examples were normalised using the applicable cost adjustments to establish a common baseline, which is then used to pro-rata labour requirements for all other ACM removal tasks based on estimated ACM volumes to be removed.

6.1.2 FREIGHT

6.1.2.1 TO AUSTRALIA FREIGHT

Freight volumes returning to Australia are based on the estimated volume of ACM to be removed. These estimates are carried out through high level quantity estimation of the area and material thickness removed. A bulking factor is then applied to this volume to account for natural bulking and packaging for freight.

6.1.2.2 FROM AUSTRALIA FREIGHT

Freight volumes leaving Australia considers PPE and Equipment required scaled by the quantity of ACM to be removed. Using the Old Trades example where a PPE and equipment list was provided, the volume of these items can be approximated based on the ACM removal volume.

Where reinstatement is required, the estimated volume is assumed to be the same as the ACM removal (before bulking factor is applied). A separate bulking factor is then applied to consider improved efficiency of packaging and dunnage. The reinstatement options presented in the Planning and Costing Tool also provide minimum material thicknesses that may be greater than those removed and therefore might result in increased reinstatement material volume.

6.1.2.3 FREIGHT RATES

Freight rates are provided by AAD and are to use the same rate for freight to/from Australia. Example rates for the 2025/26 season are provided in Table 2.

Table 2 – Season 2025/26 freight rate table

Station	Freight ($/m^3)
Casey	2092.40

Davis	3187.77
Mawson	6191.74
Macquarie Island	3780.57

6.1.3 LABOUR

Labour associated with ACM removal is estimated using the previously discussed pro-rata approach using the previously completed AAD asbestos removal projects. This pro-rata based on ACM removal volume will give the AAD an estimate of time required for removal for ACM that can then be factored against the labour rates provided in Table 3 to determine the labour cost for a given removal task.

The following rates are hourly labour rates provided by the AAD based on 2025/26 season rates. The Labour and G&S rates are to be summed to provide a total labour rate that is applied to estimated removal labour effort.

Table 3 – Season 2025/26 AAD labour rate table

Station	Ant Lbr ($/hr)	Ant G&S ($/hr)	Total Ant Lbr ($/hr)
Casey	171.95	479.99	651.94
Davis	171.95	445.99	617.94
Mawson	171.95	555.84	727.79
Macquarie Island	171.95	318.48	490.43

Reinstatement labour rates have been provided by North Projects Limited on an hour per m2 basis.

6.1.4 COST ADJUSTERS

To prevent linearly disproportionate scaling of cost, the labour costs are increased using a simple cost adjuster based on removal complexity and accessibility. For each removal task costed, the variable labour rate will be increased by the sum of the cost adjusters.

$$Labour\ Cost = [Base\ Labour\ Cost] \times (1 + \sum Adjusters)$$

6.1.4.1 COMPLEXITY ADJUSTER

The complexity adjuster reflects how straightforward or involved the removal task is based on the item being removed, the scale of the work and the associated dismantling or work interfaces. It is independent of ACM condition, friability and physical access. A small, discrete item such as a gasket is simpler than removing distributed flooring or wall linings, while foundations and other large or highly involved removal works attract a higher adjustment. The proposed complexity cost adjusters are provided in Table 4.

Table 4 – Removal complexity cost adjuster

Complexity	Adjuster	Typical removal task
Low	0%	Small, discrete component removal with minimal dismantling, or straightforward removal of a defined area with limited interfaces. Examples include gaskets, grilles, backing boards, replaceable components, flooring, wall linings, ceiling linings or cladding.
Medium	10%	More involved or extensive removal of flooring, wall linings, insulation, cladding or service-related items requiring staged dismantling or coordination with other building elements.

High	20%	Large or highly involved removal works, such as ACM associated with foundations, substantial building fabric, major plant interfaces or extensive dismantling and sequencing.

6.1.4.2 ACCESSIBILITY ADJUSTER

The accessibility adjuster accounts for factors that may reduce removal efficiency, including physical access constraints, work in occupied or operational areas, the need for access equipment or isolation, and the condition and friability of the ACM. The material risk rating from the Asbestos Register forms part of this assessment, with ratings of 3–4 generally associated with easier removal, a rating of 2 with more challenging removal, and a rating of 1 with extreme removal conditions. The final adjuster is selected based on the overall circumstances of the removal task. The proposed accessibility cost adjusters are provided in Table 5.

Table 5 – Accessibility risk for ACM removal cost adjuster

Accessibility	Adjuster	Indicative Material Risk	Accessibility Examples
Easy	0%	3–4	• Gasket replacements not affecting operations. • Electrical backing boards. • Component replacements, such as grilles and doors. • Internal wall linings in typically unoccupied spaces. • Readily accessible floor linings.
Challenging	20%	2	• Gasket replacements affecting operations. • Floor sheeting requiring additional controls or coordination. • Internal wall linings in occupied spaces. • Exterior wall cladding. • Buried or partially buried items, such as pipework. • Loose or damaged material requiring additional controls.
Extreme	40%	1	• Work in confined spaces, such as subfloors or roof spaces. • Work at height, including ceilings, soffits and roofing. • Items requiring specialist access or significant operational isolation. • Friable or significantly degraded ACM requiring extensive containment and handling controls.

6.1.5 INFLATION ADJUSTER

Inflation is considered in the forecasted costing to improve the accuracy of the future ACM removal costs. The tool has built in ability to adjust the inflation rate as required. A default value of 3.5% is used for the 10-year plan but can be adjusted within the tool.

$$Future\ Cost = [Cost\ today] \times (1 + r)^n$$

The estimated inflation rate ‘r’ is taken as 3.5% and entered at 0.035 in the above equation. The value ‘n’ is the number of years from the current year.

6.1.6 REINSTATEMENT MATERIAL COST

Reinstatement material rates have been provided by North Projects Limited on a $/unit rate. The variance is required for the different types of reinstatement options from individual gaskets to vast areas of wall lining. The units and scale are driven by the unfactored volumes and quantities calculated for the ACM removal.

7 CONSTRAINTS

The ARP is a fit-for-purpose, high-level planning tool. Its outputs remain subject to a range of constraints that may influence seasonal planning, resourcing and delivery, as summarised below:

- **Station resourcing and competing priorities**
  Asbestos removal is constrained by the availability of suitably trained personnel and supporting trades at each station. These resources must be coordinated with maintenance, capital works, operational support and other planned station activities, which may limit the labour available in a given season.
- **Accommodation and bed capacity**
  Station bed capacity limits the number of additional personnel who can be deployed for asbestos removal and reinstatement. The tool provides a high-level indication of bed demand, but actual allocation remains subject to seasonal population planning and competing operational requirements.
- **Logistics, freight and waste return**
  Removal depends on sufficient freight capacity for personnel, equipment, PPE, containment materials and replacement products, together with the return of packaged ACM waste. Freight or on-station storage limitations may require activities to be deferred, divided across seasons or combined with other planned works.
- **Seasonal access and environmental conditions**
  Delivery is constrained by station access windows, transport schedules, weather, temperature and daylight. These factors are particularly relevant to external work, work at height and activities outside the station thermal envelope.
- **Station operations and project interfaces**
  Removal may require temporary isolation of occupied areas, critical services, plant or access routes, potentially affecting station operations and science activities. Works must therefore be coordinated with maintenance shutdowns, asset renewals, demolition, station upgrades and other planned projects.
- **Removal complexity, accessibility and reinstatement**
  Concealed materials, restricted spaces, work at height, adjacent services and ACM incorporated into building or structural elements may reduce productivity or prevent complete removal. The need for enabling works and reinstatement may also govern the timing, cost and practical extent of removal.
- **Source information and programme uncertainty**
  Programme outputs depend on the accuracy and currency of the Asbestos Registers, quantity estimates, condition information, maintenance plans and asset end-of-life dates. Changes or gaps in these inputs may affect prioritisation, removal timing, labour requirements, freight volumes and forecast costs.

8 DAVIS STATION ARP

The Davis programme was developed by applying the prioritisation approach in Section 4 and the programme assumptions in Section 3.6. On this basis, the Davis programme includes 26 ACM entries. Two are recorded as complete, leaving 24 active entries. Nine entries have been assigned a removal year: four in 2028, two in 2029, one in 2031 and two in 2032. The 2028 and 2032 works are mainly linked to asset end of life, with an additional high-risk item included in 2028, while the 2029 and 2031 works reflect priority assessment outcomes and opportunities to coordinate removal with planned station works.

The remaining 15 active entries are managed under the AMP without an assigned removal year. These include six gasket entries linked to future maintenance, six hut entries where removal is not currently preferred, and three other items for which ongoing inspection, encapsulation or management remains the more practical approach. These items should be reconsidered when asset plans, maintenance timing, condition information or practical removal opportunities change. Figure 2 shows the planned future ACM removal volume for Davis station.

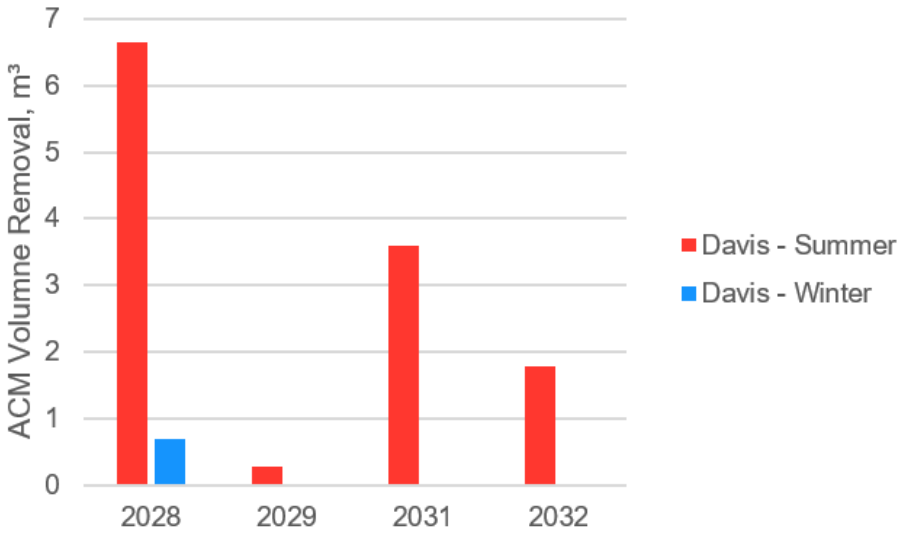
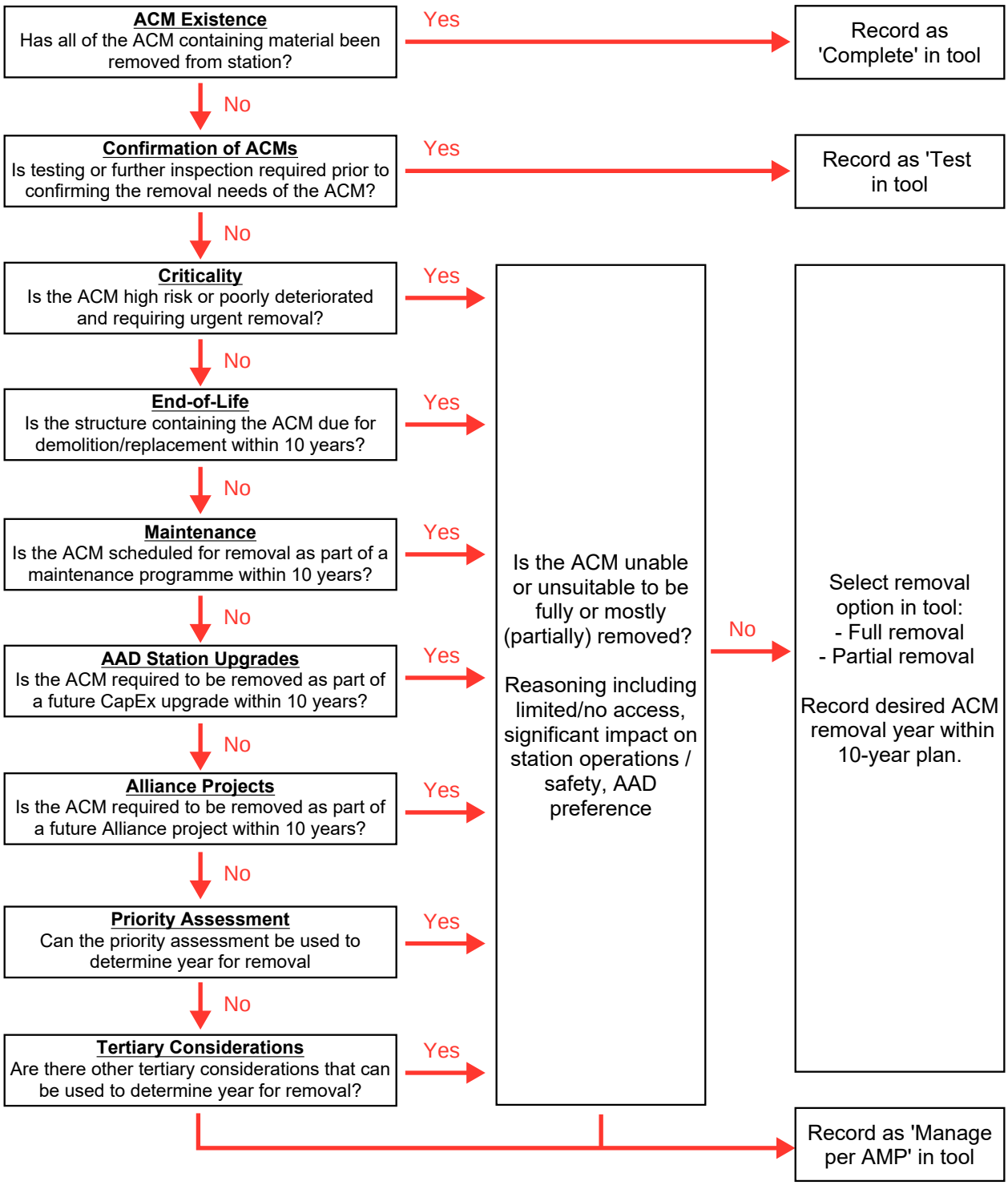


Figure 2 – Planned Future ACM Removal Volume for Davis Station

9 APPENDIX

9.1 APPENDIX A – PRIORITISATION DIAGRAM

AAD - Decision Making Flow Chart for Scheduling Removal of ACM's from Station



9.2 APPENDIX B – COSTING ASSUMPTIONS

The following are the costing assumptions and limitations as provided by North Projects Ltd.

9.2.1 GENERAL BASIS

- All rates are in Australian dollars and exclude GST.
- Refer to the “Comments” column for item-specific qualifications, assumptions and exclusions.
- Rates are preliminary planning allowances and are not for funding application purpose
- Freight, special packaging, handling, storage and shipment to Antarctica are excluded from material rates
- Proprietary and specialist items are subject to confirmation by suppliers and contractors.
- No allowance has been made for engineering design, shop drawings, certification.
- Rates exclude any overheads and profit, contingency, escalation
- No allowance for minimum-order quantities

9.2.2 PRODUCTIVITY AND LABOUR

- Labour production rates represent direct person-hours at the workface under normal, reasonably continuous Australian construction conditions.
- Production rates assume competent and appropriately licensed personnel, suitable tools, a prepared workface and reasonable continuity of work.
- Antarctic station-specific accessibility and complexity adjustments are excluded from the base production rates
- Minimum labour hours apply per discrete work lot or location and are not additional to calculated production hours. Labour hours are to be calculated as the greater of the production-rate allowance or the applicable minimum-hours allowance.
- No allowance has been made for time lost due to station inductions, on-site training, RDOs, inclement weather, permit delays, restricted working hours, competing station priorities or other non-productive time.
- No allowance has been made for reduced productivity caused by small or fragmented work lots, restricted access, occupied areas

9.2.3 MOBILISATION AND LOGISTICS

- Mobilisation and demobilisation of personnel, tools and equipment are excluded.
- Expeditioner pre- and post-deployment training, medical assessments, travel, personnel transfers, accommodation, station beds, return-to-Australia arrangements and associated logistics are to be provided or allowed for by others.
- No allowance has been made for stockpile or storage management, unloading, internal station transport, double handling or transferring materials from storage areas to the workface.
- No allowance has been made for temporary facilities, laydown areas, weather protection.
- No allowance has been made for specialist temporary access systems, including scaffolding, EWPs, harness systems or temporary working platforms.
- No allowance has been made for any hand tools, plant and equipment.
- No allowance has been made for biosecurity.
- No allowance for spare materials.

9.2.4 REINSTATEMENT SCOPE

- Rates exclude major replacement of framing, substrates, insulation and adjoining finishes etc

- No allowance has been made for concealed deterioration, corrosion, water damage, mould, rot, structural defects or unsuitable substrates discovered after ACM removal.
- Minor fixings, consumables and stated wastage are included only where identified in the material build-up.
- Trims, flashings, skirtings and similar perimeter components are based on average allowances.
- Painting allowances assume standard surface preparation and the stated number of coats. Specialist coating system is not allowed
- Drying, curing and return visits are reflected only through the applicable production rate and minimum-hours allowance; extended delays caused by environmental conditions are excluded.

9.2.5 SERVICES AND SPECIALIST WORK

- Service isolation, disconnection, relocation, reconnection, testing and commissioning are excluded.
- Shutdown planning, operational disruption, temporary services are not considered
- No allowance has been made for modification or replacement of any building services impacted by asbestos removal or reinstatement works
- Fire doors, fire protection and penetration seals are subject to confirmation of the required fire-resistance level and selection of a compliant tested system.

9.3 APPENDIX C – DAVIS STATION PROGRAMME EXAMPLE

Australian Government
Department of Climate Change, Energy, the Environment and Water
Australian Antarctic Division

AUSTRALIAN ANTARCTIC PROGRAM

Programme Summary by User Selection

Total Freight to Antarctica	**7.85**	m^3
Total Freight from Antarctica	**19.54**	m^3
Total Labour Required	**117.50**	Total Days
Total Present Cost	**$ 486,413**	(AUD)

Station

- Casey
- Davis
- Macquarie
- Mawson

Est. Year for Wor...

- 2028
- 2029
- 2031
- 2032
- (blank)
- 2026
- 2027
- 2035
- 2040

Station	Location & Maximo Number	Item	Requirement for Removal	Treatment Type	Season	Est. Year for Works	ACM Material Type	Comments - ACM Removal Scope	Reinstatement Option	Total Freight From Ant (m3)	Total Freight To Ant (m3)	Total Labour (days)	Inflation adjusted cost, $
Davis	Emergency Power House	Internal walls (photo L108901/ 02/	End-of-life	Removal - Full	Summer	2028	Wall Cladding/Lining	Wall linings to be		1.476	0.138	5	$ 19,228
Davis	Emergency Power House	Exhaust Manifolds	End-of-life	Removal - Full	Summer	2028	Lagging, coatings			0.468	0.044	0	$ 1,899
Davis	Emergency Powerhouse	External panels & roofing panels	End-of-life	Removal - Full	Summer	2028	Wall Cladding/Lining			8.04	0.752	0	$ 32,623
Davis	Sleeping and Medical	Main plant room, asbestos coating on	High Risk	Removal - Full	Winter	2028	Lagging, coatings	Remove asbestos	Fire Protection	1.05	0.939	20.5	$ 72,523
Davis	Main Power House (MPH)	All wall sheets up to 2.2m height	End-of-life	Removal - Full	Summer	2032	Wall Cladding/Lining	Sampled tested for		2.651	0.248	0	$ 12,343
Davis	Main Power House (MPH)	All gaskets.	End-of-life	Removal - Full	Summer	2032	Gaskets			0.041	0.001	0	$ 191
Davis	Site Services L4759 J1 - North Central Junction	Potential asbestos containing material was identified on the base of the concrete stairs on the bridge over the site services that is adjacent to service Junction 1	Priority Assessed	Removal - Full	Summer	2029	Other	Remove encapsulated ACM formwork from beneath the concrete stairs.		0.06	0.006	1.5	$ 4,363
Davis	General Workshop L1118	Internal wall sheets (photo L111801).	Priority Assessed	Removal - Full	Summer	2031	Wall Cladding/Lining	Remove internal ACM wall sheets, including temporary relocation of obstructing equipment.	Wall Int. – Cold Porch Lining	5.4	5.689	89	$ 337,633
Davis	Site Services L1132	Old form work around the foundations (photo L113201).	Priority Assessed	Removal - Partial	Summer	2029	Other	Excavate locally and remove accessible ACM formwork from the concrete footings.		0.357	0.034	1.5	$ 5,610